

Self-Learning: NoSQL Data Models — DBMS

Definition: NoSQL (Not Only SQL) refers to non-relational database management systems designed for horizontal scalability, high-velocity big data, flexible schema-less data models, and high availability.

1. Detailed Technical Explanation

NoSQL databases trade traditional ACID strict transactional guarantees for high performance and horizontal scaling across distributed clusters (governed by the **CAP Theorem**).

CAP Theorem:

A distributed database system can guarantee at most **two out of three** properties simultaneously:

- 1. **Consistency (C):** All nodes read the latest data at the same time.
- 2. **Availability (A):** Every non-failing request receives a non-error response.
- 3. **Partition Tolerance (P):** System continues to operate despite network communication failures.

2. The Four Major NoSQL Data Models

NoSQL Category	Data Model	Key Features	Popular Industry Tools	Use Cases
Key-Value Store	Hash Table (Key -> Value Blob)	Fast lookup by key, ultra-high performance.	Redis, Amazon DynamoDB, Riak	Caching, session management, user shopping carts.
Document Store	JSON / BSON / XML Documents	Schema-free nested documents, rich indexing.	MongoDB, CouchDB	Content management, e-commerce product catalogs.
Column-Family Store	Sparse tables indexed by Rows & Column Families	Optimized for heavy write throughput and analytical queries.	Apache Cassandra, HBase	Time-series data, IoT analytics, financial logs.
Graph Database	Nodes, Edges (Properties)	Graph traversal queries for interconnected data.	Neo4j, Amazon Neptune	Social networks, fraud detection, recommendation engines.

3. SQL vs NoSQL Architecture Comparison

Feature	Relational SQL Databases	NoSQL Databases
Data Schema	Fixed, predefined rigid schema.	Dynamic, schema-less / flexible.
Scaling	Vertical scaling (Scale-up: bigger CPU/RAM).	Horizontal scaling (Scale-out: sharding across commodity nodes).
Transactions	Strict ACID Compliance.	BASE Model (Basically Available, Soft-state, Eventual consistency).
Joins	Native SQL JOIN operations.	Denormalized data / Application-side joins.

4. Quick Recall Flow

NoSQL -> Scale-Out Horizontal Architecture -> CAP Theorem Tradeoffs -> Key-Value | Document | Column-Family | Graph